Single-Cell Transcriptomic Landscape of the Neuroblastoma Tumor Microenvironment

Dataset: GSE253865 · Reference: Bonine et al. 2024, Cell Reports

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Quality Control Metrics

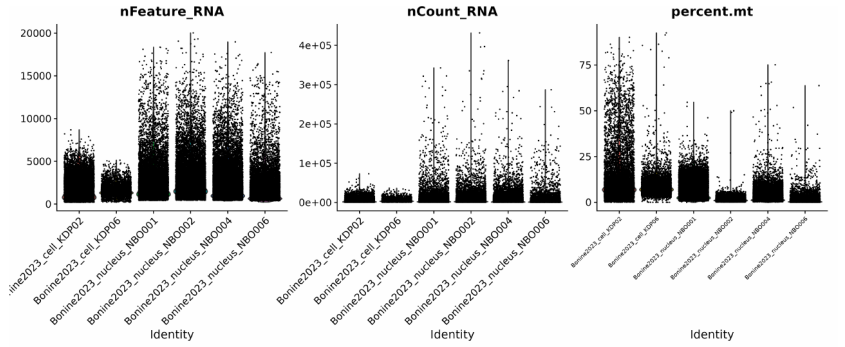


Figure QC. Violin plots showing per-cell QC metrics across all 6 samples before filtering. *Left:* Number of detected genes (nFeature_RNA). *Centre:* Total UMI counts (nCount_RNA). *Right:* Mitochondrial gene percentage (percent.mt). Cells were retained if: 500 < nFeature < 6,000; 1,000 < nCount < 50,000; percent.mt < 10%. This removed 28% of barcodes (72,652 → 52,342 cells).

Sample Summary

Sample	Assay Type	Raw Cells	Raw Genes
cell_KDP02	single cell	8540	28432
cell_KDP06	single cell	7106	24045
nucleus_NBO001	single nucleus	11309	40182
nucleus_NBO002	single nucleus	18016	42093
nucleus_NBO004	single nucleus	16081	40879
nucleus_NBO006	single nucleus	11600	36090

Figure A — Sample Overview

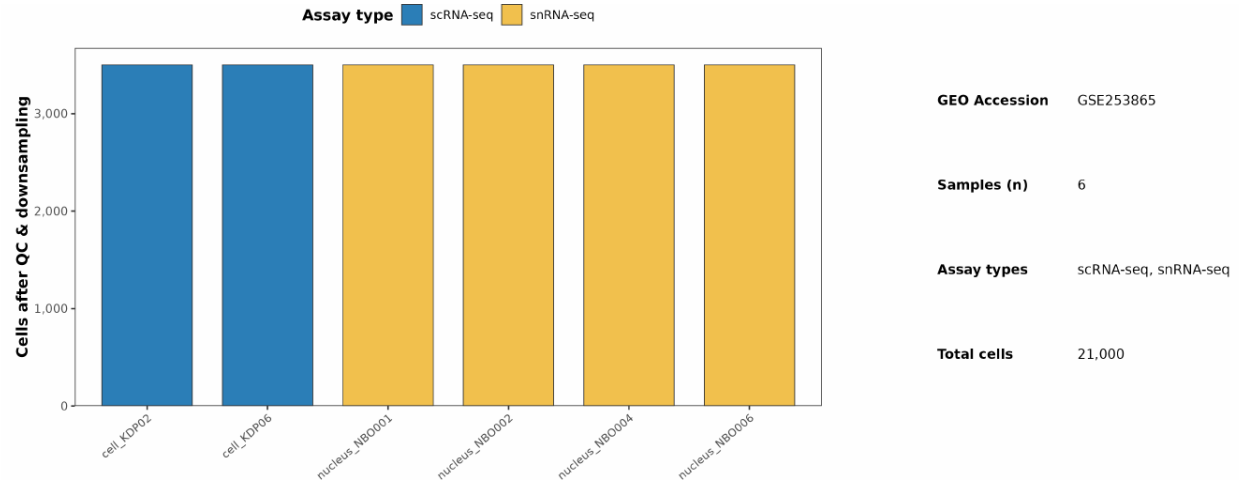


Figure B — UMAP Embeddings

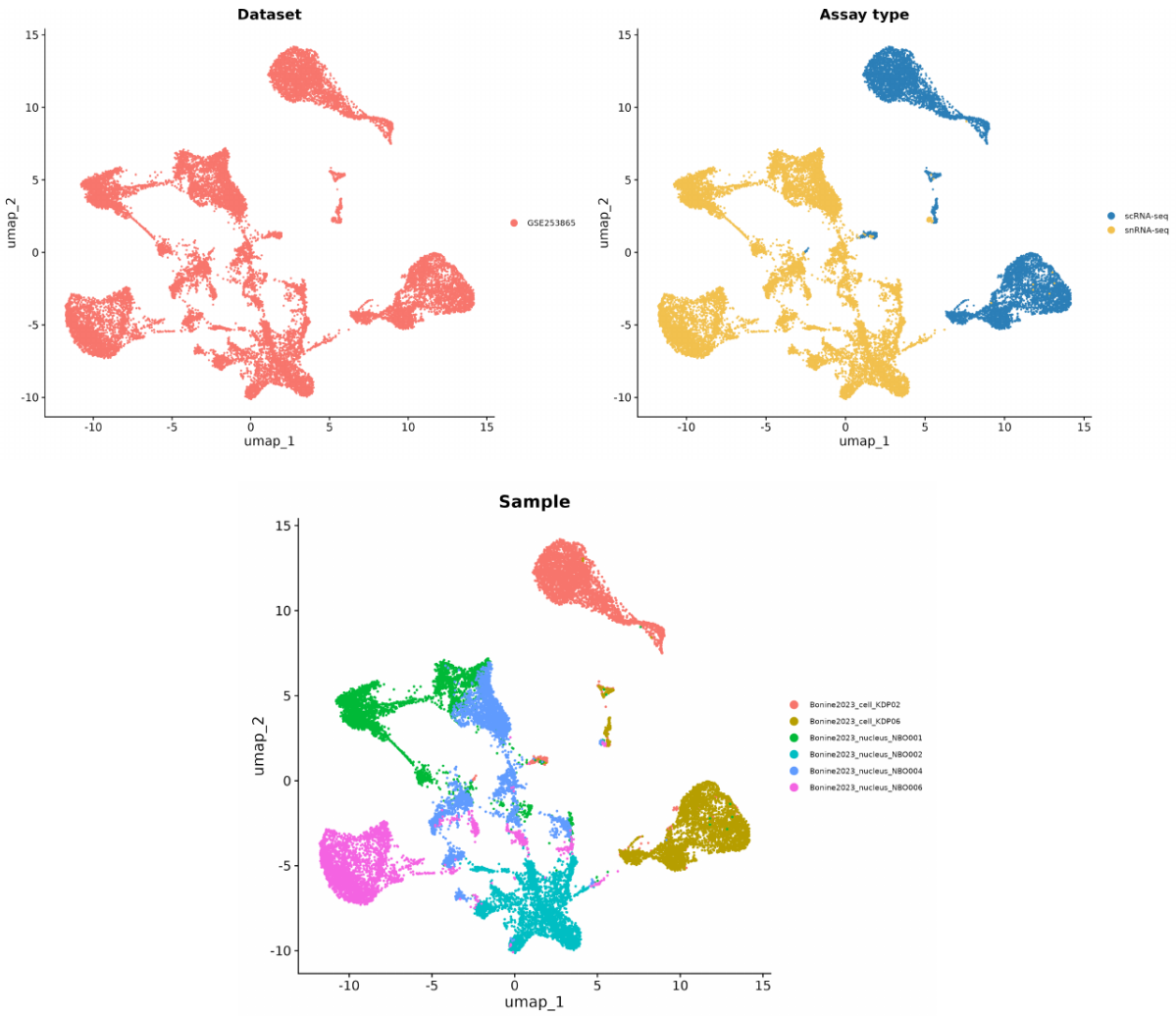


Figure B. UMAP embeddings (dims 1–20, 30 PCs) coloured by metadata. *Top-left:* Dataset of origin (all GSE253865). *Top-right:* Assay type — scRNA-seq (blue) vs snRNA-seq (yellow); note partial separation reflecting technical differences between assays. *Bottom:* Individual sample identity. Cells from different samples intermix substantially, suggesting shared biology across patients. This panel mirrors Figure 1B of Bonine et al. 2024.

Figure C — Annotated Cell Types

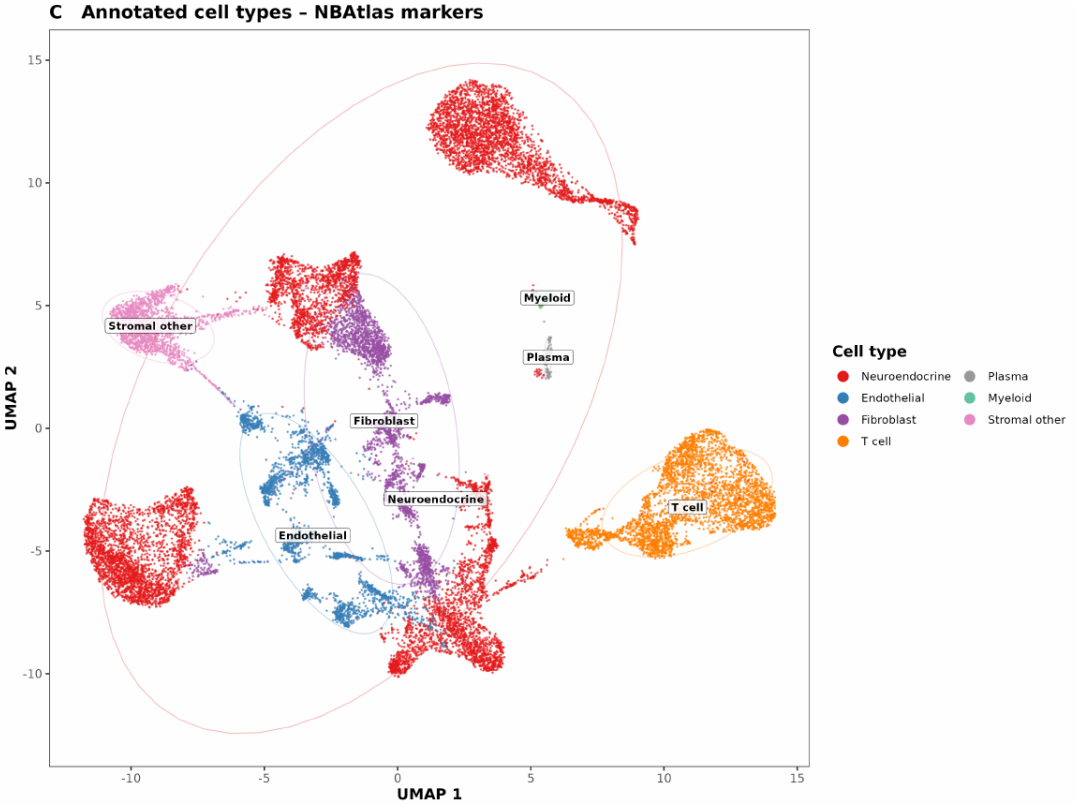


Figure C. UMAP coloured by NBAtlas-style marker-based cell type annotation. Cluster identities were assigned by computing AddModuleScore for 3-gene panels per cell type (from Bonine et al. 2024 Supplementary Table 1) and assigning each cluster to the highest-scoring type. Ellipses (75% confidence, normal distribution) and centroid labels aid visual interpretation. Seven of 12 NBAtlas cell types were detected: Neuroendocrine (48.4%), T cell (16.2%), Fibroblast (13.8%), Endothelial (12.3%), Stromal other (8.0%), Plasma (0.7%), Myeloid (0.6%). Five types (Schwann, NK cell, B cell, pDC, RBCs) were absent, likely due to the subset of samples analysed. This panel mirrors Figure 1C of Bonine et al. 2024.

Cell type composition

Figure D — Canonical Marker Gene Expression

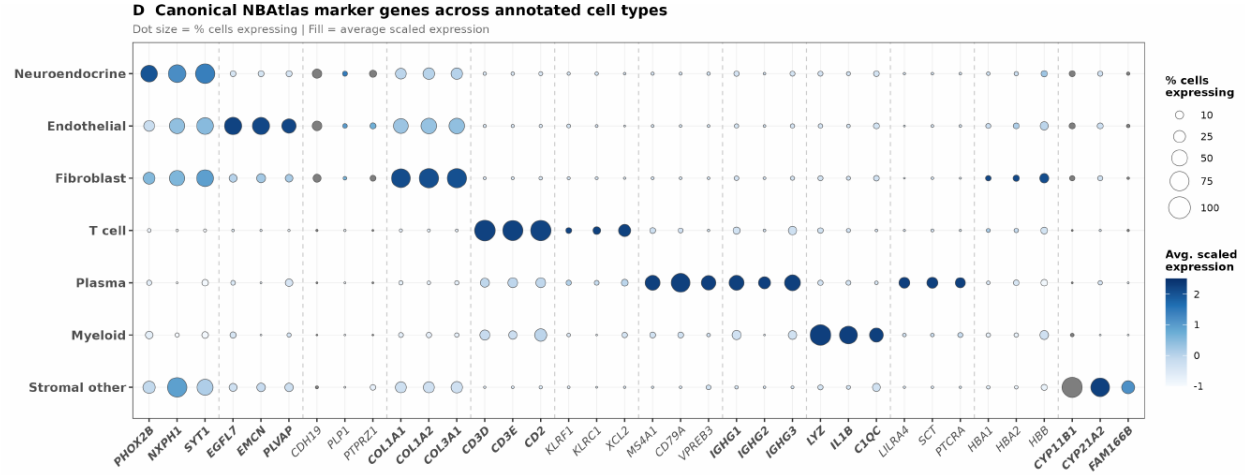
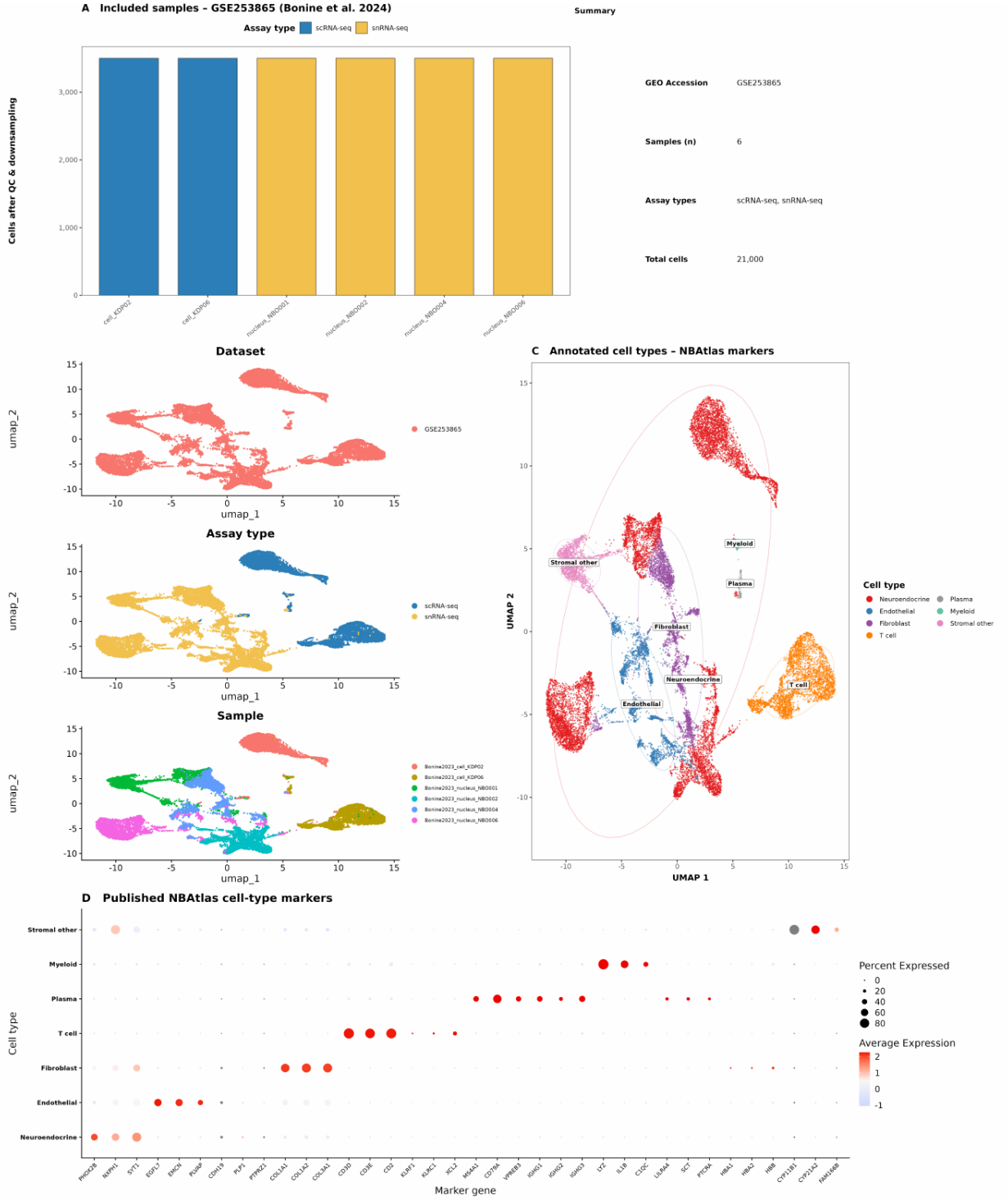


Figure D. Dot plot of canonical NBAtlas marker genes across annotated cell types. **Dot size** encodes the percentage of cells in each type expressing the gene (≥1 UMI). **Fill colour** encodes average scaled expression (white → dark blue = low → high). Dashed vertical lines separate marker gene groups by cell type. Genes are ordered by cell type (left to right): Neuroendocrine (PHOX2B, NXPH1, SYT1), Endothelial (EGFL7, EMCN, PLVAP), Fibroblast (COL1A1, COL1A2, COL3A1), T cell (CD3D, CD3E, CD2), Plasma (IGHG1, IGHG2, IGHG3), Myeloid (LYZ, IL1B, C1QC), Stromal other (CYP11B1, CYP21A2, FAM166B). Markers for Schwann, NK cell, B cell, pDC, and RBCs are shown but their cognate cell types were not detected in this subset. Neuroendocrine markers show the highest specificity and expression, consistent with the tumour-dominant composition of neuroblastoma samples. This panel mirrors Figure 1D of Bonine et al. 2024.

Summary Panel (Figures A–D Combined)



Combined Figure ABCD. All four main panels assembled in NBAtlas paper style. Top: Figure A (sample summary). Middle-left: Figure B (UMAP overview). Middle-right: Figure C (annotated cell types). Bottom: Figure D (marker dot plot). This layout directly mirrors Figure 1 of Bonine et al. 2024 (Cell Reports).